This online appendix is provided as supplementary material and will not be included for publication.

A Benchmark Stopping criteria

A.1 Benchmark stopping criteria for SOEAs

tolfun makes a stopping decision based on the fitness values. At iteration t, the minimum and maximum objective values among the λ offspring are defined as: $F^t_{\min} = \min_{i\in\{1,\dots,\lambda\}} f(x^t_i)$ and $F^t_{\max} = \max_{i\in\{1,\dots,\lambda\}} f(x^t_i)$, respectively. A set of historical minimum objective values over the most recent l iterations is defined as $\mathcal{H}^t = \{F^t_{\min}, \dots, F^{t-l}_{\min}\}$, where $l = 10 + \lfloor 30n/\lambda \rfloor$, and n is the dimension of the search space. The search stops when both $F^t_{\max} - F^t_{\min} < \epsilon_{tfun}$ and $\max \mathcal{H}^t - \min \mathcal{H}^t < \epsilon_{tfun}$ are met.

tolfunhist stops the search when $\max \mathcal{H}^t - \min \mathcal{H}^t < \epsilon_{thist}$. Different from `tolfun`, it ignores the condition on the difference between the minimum and maximum objective values.

tolflatfitness. Define $F^t_{\lfloor \alpha\lambda \rfloor}$ as the α-th percentile of the objective values among the λ offspring at iteration t. The tolflatfitness criterion stops the search when $F^t_{\min} = F^t_{\lfloor 0.75\lambda \rfloor}$ for T_{tflat} consecutive iterations. By default, the $T_{\text{tflat}} = 1$ in `pycma`. Therefore, the search is stopped immediately when $F^t_{\min} = F^t_{\lfloor 0.75\lambda \rfloor}$.

tolstagnation detects whether CMA-ES has stopped making meaningful progress. Let $\mathcal{L}$ and $\mathcal{M}$ denote the sets of minimum and median objective values among the λ offspring, respectively, recorded every five iterations. For example, when $t = 20$, $\mathcal{L}^{(t)} = \left\{F^{(5)}_{\min}, F^{(10)}_{\min}, F^{(15)}_{\min}, F^{(20)}_{\min}\right\}$, $\mathcal{M}^{(t)} = \left\{F^{(5)}_{0.5\lambda}, F^{(10)}_{0.5\lambda}, F^{(15)}_{0.5\lambda}, F^{(20)}_{0.5\lambda}\right\}$. The criterion is activated only after sufficient search history has been accumulated. In particular, it requires $t \geq l, t > n\left(5 + \frac{100}{\lambda}\right), \frac{\text{FE}-\text{FE}^{\text{last}}}{\lambda} > \frac{T_{\text{tstag}}}{2}, |\mathcal{L}^{(t)}| > 100, |\mathcal{L}^{(t)}| > 2l$, where FE^{last} denotes the number of function evaluations when the best-so-far objective value was last updated, and $l = \max\left\{\frac{T_{\text{tstag}}}{10}, \frac{|\mathcal{L}^{(t)}|}{10}\right\}$. Once these conditions are met, the criterion compares the historical and recent behavior of both the minimum and median offspring objective values, and stops when they have remained essentially unchanged for a sufficiently long period, indicating search stagnation.

A.2 Benchmark stopping criteria for MOEAs

MGBM [18] monitors the improvement between two consecutive populations using the mutual domination rate (MDR), $\text{MDR}\left(\mathcal{P}^t, \mathcal{P}^{t-1}\right) = \frac{\left|\Delta\left(\mathcal{P}^{t-1}, \mathcal{P}^t\right)\right|}{|\mathcal{P}^{t-1}|} - \frac{\left|\Delta\left(\mathcal{P}^t, \mathcal{P}^{t-1}\right)\right|}{|\mathcal{P}^t|}$, where $\Delta(\mathcal{A}, \mathcal{B})$ denotes the subset of solutions in $\mathcal{A}$ that are dominated by at least one solution in $\mathcal{B}$. A larger MDR indicates greater improvement

from iteration $t-1$ to t. Since the observed MDR is assumed to be affected by Gaussian noise, MGBM applies a Kalman filter to sequentially estimate the underlying MDR. The search is terminated when this estimated improvement falls below a predefined threshold $\hat{I}_{\min}$. Thus, MGBM detects convergence by identifying when the estimated improvement between consecutive populations becomes sufficiently small.

Entropy-based Stopping Criterion (ESC) [19] detects convergence by measuring the dissimilarity between two consecutive populations, $\mathcal{P}^{t-1}$ and $\mathcal{P}^t$, in the objective space. For an objective problem with m objectives, the objective space is partitioned into n_b^m grid cells. Let $\mathcal{C}_j$ denote the jth cell and μ the population size. The probability associated with cell $\mathcal{C}_j$ at iteration t is estimated as $p_j^t = \frac{n_j^t}{\mu}$, where n_j^t is the number of objective vectors from $\mathcal{P}^t$ falling into $\mathcal{C}_j$. ESC then measures the dissimilarity between the distributions induced by $\mathcal{P}^{t-1}$ and $\mathcal{P}^t$. The search is terminated when this dissimilarity remains unchanged for a predefined number of iterations, n_s. Thus, ESC identifies convergence when the distribution of the population in the objective space becomes stable.

ϵ-Progress-based Stopping Criterion (ϵSC) [6] monitors search progress using an ϵ-box dominance archive. The objective space is partitioned into hyper-boxes of side length ϵ, and solutions are compared according to the boxes they occupy. For example, given two objective vectors $\mathbf{u} = (u_1, \dots u_m)^\top$ and $\mathbf{v} = (v_1, \dots v_m)^\top$, $\mathbf{u}$ is said to ϵ-box-dominate $\mathbf{v}$ if $\left\lfloor \frac{\mathbf{u}}{\epsilon} \right\rfloor \prec \left\lfloor \frac{\mathbf{v}}{\epsilon} \right\rfloor$, or $\left\lfloor \frac{\mathbf{u}}{\epsilon} \right\rfloor = \left\lfloor \frac{\mathbf{v}}{\epsilon} \right\rfloor$ and $\left\| \mathbf{u} - \left\lfloor \frac{\mathbf{u}}{\epsilon} \right\rfloor \right\| < \left\| \mathbf{v} - \left\lfloor \frac{\mathbf{v}}{\epsilon} \right\rfloor \right\|$ is satisfied. The archive retains representative non-dominated solutions from the occupied ϵ-boxes, providing a coarse approximation of the Pareto front. Let c_ϵ denote the progress counter. It is increased whenever a newly generated solution is not dominated by the current archive and lies in a previously unoccupied ϵ-box. The search is terminated if c_ϵ is not updated for T_ϵ consecutive iterations. Thus, ϵSC detects convergence when the search no longer discovers new non-dominated regions of the objective space.

B Additional Experimental Results

Here, we provide additional results for the experimental settings omitted from the main text for brevity. For SOEA, we report the normalized cost-aware performance across the 24 `bbob` functions for dimensions $n = 3$ and $n = 5$ in Figure B.1. For MOEA, we report the corresponding cost-aware performance across DTLZ1-DTLZ7 for $m = 2$ and $m = 5$ in Figure B.2.

C Ablation Study of Elitist Prediction Condition

Recall that the prediction model includes an elitist prediction constraint,

$$g_{\text{comb}}(t+1 \mid \boldsymbol{\xi}) \leq B_t,$$

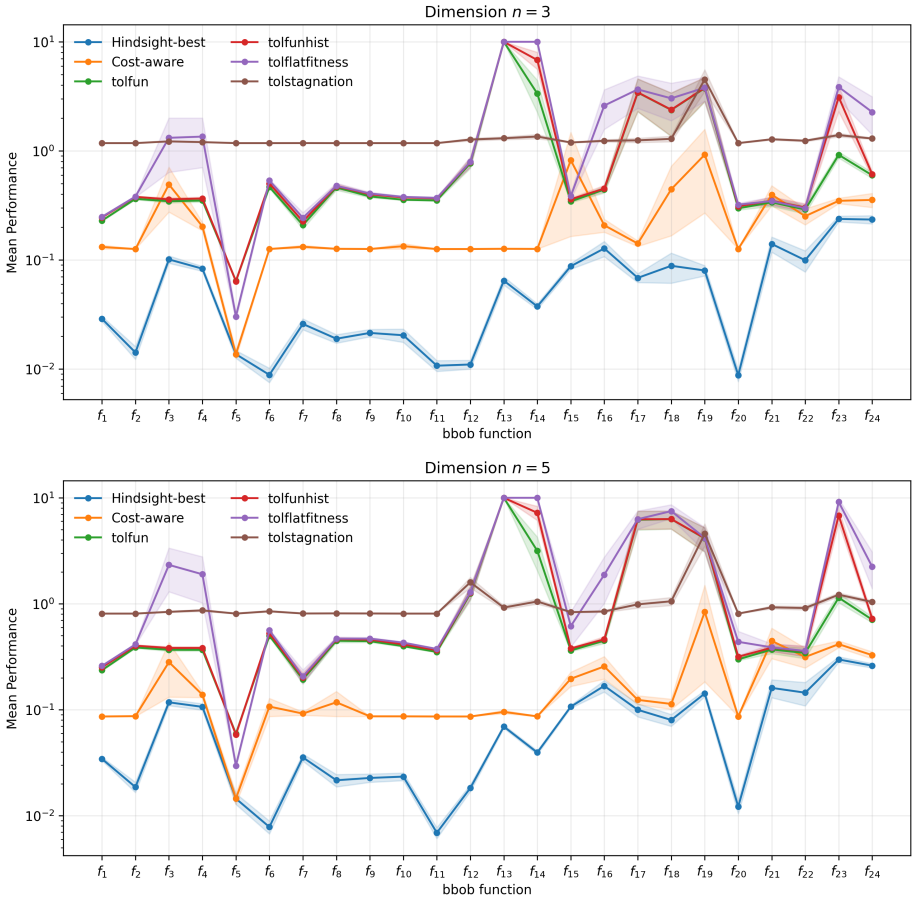


Fig. B.1: SOEA mean normalized cost-aware performance of the seven stopping methods across the 24 `bbob` functions for dimensions $n = 3$ and $n = 5$. For each function and stopping method, the solid line shows the mean performance over 15 problem instances, and the shaded region represents $\pm$ one standard error. Lower performance indicates better stopping performance.

as given in Constraint (5). This constraint ensures that the predicted next best-so-far objective value is no worse than the current best-so-far value. We conduct two ablation studies to assess the effect of this constraint on prediction accuracy and stopping performance.

C.1 A prediction example without the elitist prediction constraint.

We perform a same illustrative prediction experiment from Section 4 by removing the elitist prediction constraint from the learning-curve fitting procedure. The

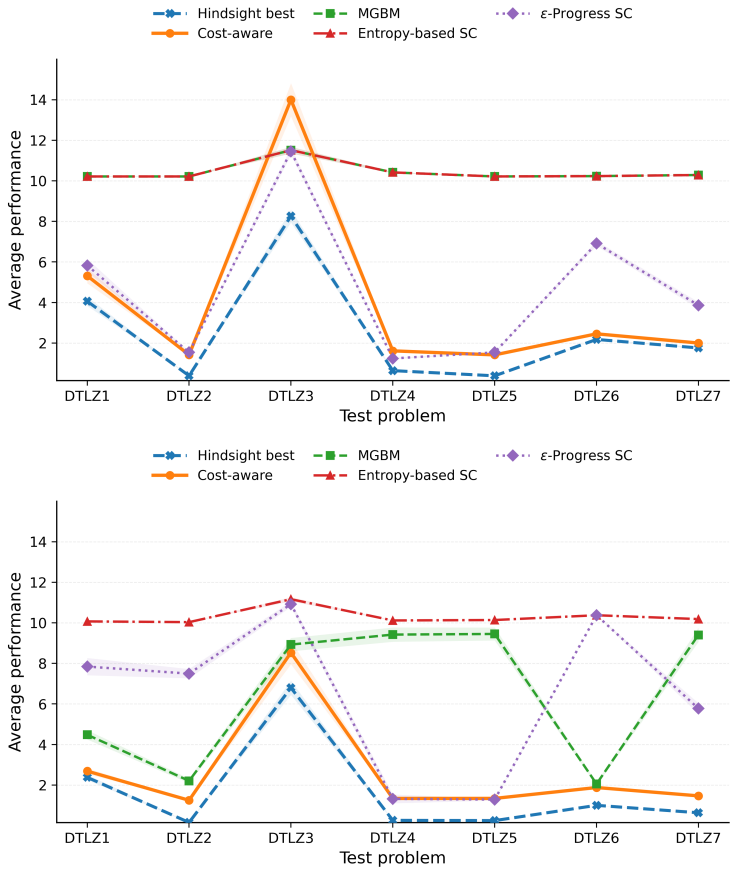


Fig. B.2: MOEA mean normalized cost-aware performance of the stopping criteria across DTLZ1–DTLZ7 problems for 2(top) and 5(bottom) objectives. The line shows the mean performance over 31 replications, and the shaded region represents ± one standard error. Lower values indicate better stopping performance.

results are shown in Figure C.3. Compared with Figure 1 with the condition, it demonstrates that the constraint improves prediction consistency. Without the constraint, after the sharp improvement around generation 26, the predicted next best-so-far values become worse than the current best-so-far value, causing a clear prediction mismatch. In contrast, the constrained model respects the non-increasing best-so-far trajectory and adapts immediately to the new objective level.

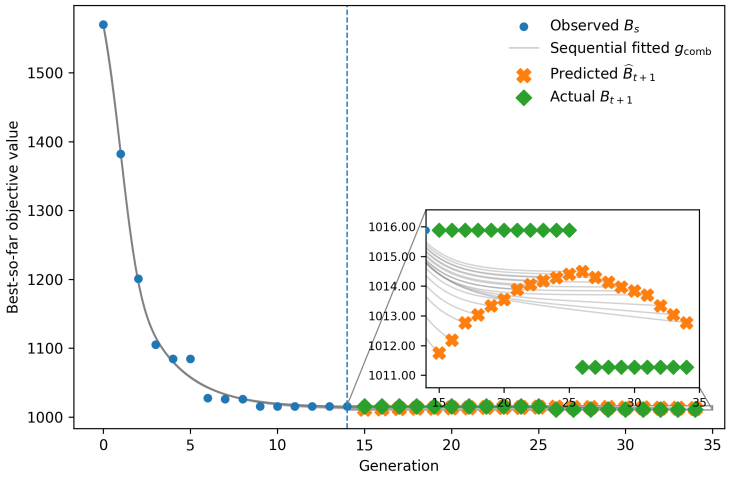


Fig. C.3: A representative prediction example for CMA-ES on one instance of f_{15} in the `bbob` suite with $n = 3$, without the elitist prediction constraint in Eq. 5. Prediction starts at generation 15 and continues for the next 20 generations. For the final fitted curve at generation 35, the model weights are $\omega_1 = 0.61$, $\omega_2 = 0.00$, $\omega_3 = 0.36$, and $\omega_4 = 0.03$.

C.2 Stopping results without the elitist prediction constraint.

To evaluate the effect of the elitist prediction constraint, we conduct ablation studies for both SOEA and MOEA. In each case, we remove the elitist prediction constraint from the learning-curve fitting procedure while keeping all other experimental settings unchanged, as described in Section 5.

SOEA. We consider CMA-ES on the 24 `bbob` functions with dimension $n = 2$, removing the constraint $\hat{L}_{t+1} \leq L_t$ while retaining the non-negativity constraint $\hat{L}_{t+1} \geq 0$. The results are reported in Table C.1. Removing the constraint slightly changes the mean performance of the cost-aware criterion from 0.22 ± 0.03 (as shown in Table 1) to 0.21 ± 0.01 and reduces the mean stopping FEs from 389.37 ± 55.09 (as shown in Table 2) to 366.20 ± 24.50.

Table C.1: Mean performance and stopping function evaluations (FEs) of the stopping criteria for CMA-ES on the 24 `bbob` functions with dimension $n = 2$, without the elitist prediction constraint. Results are reported as mean ± one standard error. Lower values indicate better performance and fewer evaluations, respectively.

Stopping criterion	**Mean performance**	**Mean stopping FEs**
`tolfun`	0.69 ± 0.08	1331.08 ± 161.62
`tolfunhist`	0.96 ± 0.11	1859.60 ± 216.48
`tolflatfitness`	1.17 ± 0.13	2282.35 ± 254.78
`tolstagnation`	1.57 ± 0.01	3081.57 ± 17.77
Cost-aware	$\mathbf{0.21} \pm 0.01$	$\mathbf{366.20} \pm 24.50$
Hindsight-best*	0.06 ± 0.01	50.40 ± 2.43

Table C.2: Mean performance and stopping function evaluations (FEs) of the stopping criteria for MOEA/D with $m = 3$ objectives. Results are reported as mean ± one standard error. Lower values indicate better performance and fewer evaluations, respectively.

Stopping criterion	**Mean performance**	**Mean stopping FEs**
MGBM	9.65 ± 0.14	9107.83 ± 106.83
Entropy-based SC	10.54 ± 0.07	10000.00 ± 0.00
ϵ-Progress SC	6.18 ± 0.29	5556.45 ± 249.24
Cost-aware	$\mathbf{3.58} \pm 0.30$	$\mathbf{1240.09} \pm 14.12$
Hindsight best	2.28 ± 0.21	874.88 ± 111.48

MOEA. We conduct the same ablation study for MOEA/D with $m = 3$ objectives by removing the constraint $\hat{R}_{t+1} \leq R_t$. As shown in Table C.2, the mean performance of the cost-aware criterion slightly worsens from 3.54 ± 0.30 (as shown in Table 3) to 3.58 ± 0.30, while the mean stopping FEs decrease from 1289.63 ± 29.83 (as shown in Table 4) to 1240.09 ± 14.12. Unlike SOEA, the standard error of performance remains unchanged in MOEA, while that of stopping FEs decreases from 29.83 to 14.12, suggesting that removing the constraint reduces variability in stopping FEs without noticeably affecting performance variability.

Removing the elitist prediction constraint leads to earlier stopping in both SOEA and MOEA experiments. It marginally improves performance in SOEA but slightly worsens performance in MOEA, none of the differences is statistically significant. Overall, while the elitist prediction constraint seems logical, these preliminary results suggest that its benefit may be small and depend on the problem characteristics such as the trade-off between solution improvement and evaluation costs.